Seminar - *Critical Topics in AI*

AGI - Benchmarks, Feasibility, Implications

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Abstract

The concept of human-level or artificial general intelligence (AGI) has been a much-anticipated technological goal for many years. Throughout the history of artificial intelligence, experts have repeatedly and incorrectly predicted AGI's short-term achievement. But despite many impactful advancements in the field of artificial intelligence, no clear path towards AGI is known decades after its inception. In this report, we demonstrate that the main problem in the development of AGI lies in its ambiguous definition. We relate the vagueness of the term AGI back to its basis in the concept of intelligence and point out how the resulting ambiguity allows for individual and subjective understandings of AGI. Additionally, we highlight the impact of this ambiguity on benchmarking strategies, technical approaches for implementation, the prognosis of development timelines, and predictions of impacts related to AGI. As a consequence of AGI's intrinsic ambiguity, no targeted development of AGI is possible without a better understanding of the underlying concept of intelligence as a whole. Nevertheless, discussions on this topic rarely address the clarification of subjective conceptions of AGI or acknowledge the lack of a universal AGI definition.

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1 Introduction

Artificial general intelligence (AGI) is a topic that has gained increasing interest in both research and public discussion in recent years. The main driver of this increasing interest was the release and widespread adoption of ChatGPT and subsequent large language models (LLMs) (Brown et al., 2020). The rapid improvement of these LLMs with techniques such as chain-of-thought reasoning (Wei et al., 2022), tool use (Schick et al., 2023), and independent, agentic action-taking has raised the question of how this advancement in the broader field of artificial intelligence (AI) relates to the concept of AGI. Several of the companies developing some of the most popular LLMs, including OpenAI (OpenAI, 2024), Google (Dragan, Shah, Flynn, & Legg, 2025), Amazon (Luan & Abbeel, 2024), and xAI (Kay, 2025), have stated AGI as a long-term goal of their scientific efforts.

Despite the recent increase in focus on AGI, the concept is not new. Concepts of machines exhibiting intelligence equivalent to, or even surpassing, that of humans can be found not only in historical science fiction but also in science, mainly since the 1950s. Since then, the belief that AGI will be brought into existence within the foreseeable future is a constantly recurring belief among many influential figures in the field of AI. This becomes evident by comparing some statements of past and present influential speakers. Both the Dartmouth workshop attendee and co-founder of MIT's AI lab, Marvin Minsky (Darrach, 1970):

> 'In from three to eight years we will have a machine with the general intelligence of an average human being.'
>
> (Marvin Minsky - 1970)

and the AI pioneer and co-founder of one of the world's first university computer science departments, Herbert A. Simon (Simon, 1965, p.96):

> '[...] machines will be capable, within twenty years, of doing any work a man can do.'[1]
>
> (Herbert A. Simon - 1965)

predicted AGI to become a reality within a few years in the past. From today's perspective, the predictions of these two men, who were awarded the Turing Award for their foundational work in AI in 1969 and 1975, respectively, have proven incorrect. Yet they have a strong resemblance to recent predictions like the ones by DeepMind co-founder Demis Hassabis (Rose, 2025):

[1] It is worth noting that Simon described his own prediction as being the result of an 'armchair analysis' that he conducted in the context of estimating the impact of machine-based automation on human labor, and not a detailed scientific analysis.

> 'We'll have something that will exhibit all the cognitive capabilities humans have, maybe in the next five to 10 years.'
>
> (Demis Hassabis - 2025)

or OpenAI co-founder and CEO Sam Altman (Altman, 2024):

> 'It is possible that we will have superintelligence in a few thousand days'[2]
>
> (Sam Altman - 2024)

While these past and present predictions are similar, the circumstances in which they were made differ significantly. Not only have the technical foundations of AI evolved, but so have other scientific fields adjacent to the concept of intelligence, as well as real-world experiences with AI use and its implications.

As part of this seminar, which discusses critical aspects of AI technologies and their use, we investigate the current state of AGI in this report. Instead of approaching the topic from a technical direction, we put our main focus on the ambiguity that is intrinsic to AGI as a research field, and is mainly responsible for the recurring predictions of near-term AGI existence. To do so, we start by introducing the definition of AGI and its current vagueness in Section 2. We then investigate the impact of this ambiguity on recently proposed AGI benchmarks (cf. Section 3), technical approaches (cf. Section 4.2), and estimation of AGI development time scales (cf. Section 4.1), as well as the prediction of implications associated with AGI (cf. Section 5). While the ambiguous definition of AGI is the concern that we built this report around, we also present up-to-date examples from AGI as a scientific field and the broader discussion around it to highlight the current state of AGI. Finally, we provide an outlook on the future development of AGI (cf. Section 6), taking into account the impact of its ambiguous foundations on current research.

2 Definition

To better understand the definition of AGI and its intrinsic ambiguity, it can help to break the term into its three parts (Russell & Norvig, 2021):

1. **Artificial:** The first part of the AGI acronym is the easiest to understand, because it aligns with the intuitive understanding of the term 'artificial'. In contrast to its natural counterpart, artificial intelligence is non-biological or human-made.

[2] Altman's prediction here is for artificial superintelligence (ASI) that, by his own definition, surpasses AGI. He is criticized for the vague formulation of 'few thousand days'.

2. **General:** While the meaning of this term does not align as directly with its intuitive understanding, its meaning in the definition is still well defined. It represents the assumption that AGI is equivalent to human intelligence in a broad spectrum or even all possible tasks. The difficulty posed by this part of the definition is not its meaning, but ways to measure and confirm it, which we further discuss in Section 3.

3. **Intelligence:** The appearance of the term 'intelligence' as part of the phrase 'artificial general intelligence' is the origin of the ambiguity that impacts all work and discussions in the field of AGI. As intelligence, as a general concept, is not sharply and well defined, the vagueness of its definition is transferred to the definition of AGI.

Since intelligence is central to human self-understanding, it is part of a multitude of scientific disciplines. These include, but are not limited to, neuroscience, psychology, biology, philosophy, as well as mathematics, and in the case of AI, even computer science. Each of these fields deals with different aspects and parts that are either considered aspects of intelligence itself or closely related to it. Examples of these are rationality, the importance of external behavior and internal thought processes, reasoning capabilities, creativity, the ability to learn and abstract, as well as self-awareness and emotions. As a result, intelligence is defined differently depending on the scientific field in which it is discussed, and on top of that, it is considered an incompletely understood concept in most of those fields (Hutson, 2025). While this diverse understanding is not unusual for a broad subject like intelligence, it poses a problem for AGI because it prevents a consensus on a universal definition of intelligence at this time.

The fact that the vaguely defined term 'intelligence' builds the foundation of AGI as a concept leads to an ambiguous and subjective understanding of AGI (Hendrycks et al., 2025). This ambiguity, in turn, affects all topics related to AGI. We showcase this propagation of ambiguity through different areas related to AGI in the following sections. The main actionable insight of understanding AGI's intrinsic ambiguity is that it is necessary to explicitly disambiguate any information or discussion related to AGI. To cope with this ambiguity, the question 'What underlying understanding of intelligence is assumed as the foundation of the term AGI in this context?' needs to be put first. Otherwise, misunderstandings based on different subjective interpretations of the term AGI will complicate or even prevent any discussion related to it.

That AGI is an ambiguous and overloaded term is exemplified by the recent trend to replace it with alternative formulations to better reflect the contextual subjective understanding of the term. As mentioned in Section 1, several companies have stated AGI as a development goal. However, recently, many of these companies have started to replace the ambiguous term 'AGI' with individualized formulations (Field, 2025). Examples of this include Meta's term 'personal superintelligence', Microsoft's formulation of 'humanist superintelligence', Amazon's 'useful general intelligence', and Anthropic's use of 'powerful AI'. While

the use of individualized terms to refer to AGI can help raise awareness that a subjective interpretation underlies the respective formulation, it cannot resolve the fundamental problem of AGI's intrinsic ambiguity.

3 Benchmarks

As discussed in the previous section, the definition of AGI is vague and does not offer universally accepted criteria to distinguish AGI from other forms of AI. Therefore, specialized benchmarks are necessary to assess potential AGI systems and to quantify progress in AGI development. In this chapter, we introduce and compare some popular and recent AGI benchmarking approaches.

3.1 The Turing Test

The Turing test was introduced in 1950 by Alan Turing as a technique to test a machine's ability to exhibit human-equivalent intelligence (Turing, 1950). It is based on the concept of indistinguishability in the use of language between the tested machine and a human. To test this, a human evaluator can communicate, typically in writing, with both a machine and a human. If the evaluator is unable to distinguish the machine from the human conversational partner based on the communication transcripts, the machine passes the test.

Today, the Turing test is not considered an effective measure of human-equivalent intelligence, mainly because it was passed multiple times by simple and even non-AI-based dialog systems like ELIZA (Weizenbaum, 1967). Skilled language use can be a sign of intelligence, which often uses some form of language as a tool (Fedorenko, Piantadosi, & Gibson, 2024; Riley, 2025). But several forms of language models have proven that learning to produce language does not guarantee intelligence as a byproduct.

3.2 Humanity's Last Exam

Humanity's Last Exam is a benchmark to test expert-level knowledge in LLMs (Phan et al., 2025). It consists of 2500 questions from various scientific fields that are text- and image-based. The benchmark was created to mitigate the typically very high scores of modern LLMs in previous LLM benchmarks by introducing harder questions. It was therefore created as a more modern and meaningful assessment of LLMs and is not directly intended to test for AGI traits. The benchmark's authors explicitly state that a high score in Humanity's Last Exam does not, on its own, qualify a system as AGI. Yet, the underlying concept of expert-level knowledge in not just one, but many academic fields is often considered to be at least one aspect of AGI.

3.3 Abstraction and Reasoning Corpus

Unlike Humanity's Last Exam, the Abstraction and Reasoning Corpus (ARC) is meant to be a direct benchmark for AGI (ARC-AGI, 2019). ARC-AGI focuses on fluent intelligence, which is defined by the ability to apply reasoning and generalize previously learned skills to previously unseen problems (Chollet, 2019). This distinguishes it further from Humanity's Last Exam, which tests mainly crystallized intelligence, which is characterized by learned knowledge and abstraction. ARC-AGI-2, the current version of the benchmark, uses static visual puzzles that require symbolic interpretation and subsequent reasoning to be solved efficiently (Chollet, Knoop, Kamradt, Landers, & Pinkard, 2026). The upcoming ARC-AGI-3 expands these concepts by introducing conceptually similar, but interactive puzzles (ARC Prize Foundation, 2026). The fundamental idea of all ARC-AGI versions is to create tests that are relatively simple to solve for humans but hard for current AI systems, to gradually steer systems developed for these benchmarks towards human-like intelligence.

3.4 Physical Turing Tests

The property of embodiment and the ability to interact with the physical world are often regarded as central to human intelligence and to intelligence as a general concept. As a consequence, multiple AGI benchmarks focus on physical aspects of intelligent behavior. Many of these follow the same heuristic as the Turing test and the ARC-AGI benchmark, and try to capture aspects of intelligence currently not exhibited by AI systems. To do so, these benchmarks focus on tasks that are mostly trivial for humans but hard to solve for current AI systems.

Examples of such benchmarks are the IKEA test (Turk, 2015), in which a robot system has to assemble flat-pack furniture, or the coffee test (Goertzel, Iklé, & Wigmore, 2012), in which an embodied AI is tasked to enter a standard human home and independently prepare a cup of coffee by locating and utilizing all necessary ingredients and equipment. A test that more closely resembles the core idea of indistinguishability in the original Turing test is the physical Turing test proposed by Jim Fan (Sequoia Capital, 2025). In this test, a robot has to clean up a messy living environment and prepare a dinner, but is judged by the indistinguishability of the result from human labor rather than its strategy during the task fulfillment.

3.5 The State of AGI Benchmarking

The examples of AGI benchmarks presented above highlight that there are currently many diverse understandings of what a test for AGI needs to examine. The diversity of benchmarking approaches can be traced back to the diversity of AGI interpretations discussed in Section 2. As people understand the ambiguous definition of AGI differently, their subjective views on AGI are based on

different properties, which their respective benchmark proposals are optimized to test for. The fact that most of the benchmarks are easy to combine does not solve this ambiguity issue. It would still leave the question open as to which properties have to be included in a combined benchmark and when all aspects required for AGI are sufficiently covered.

Although the Turing test is not used as an AGI benchmark today, it remains an instructive example of the difficulty of defining evaluation metrics for intelligence before developing intelligent systems. The fact that language proficiency was considered a suitable measure of intelligence before the invention of highly capable language and dialog models can be seen as an indicator that many of the concepts underlying modern AGI benchmarks might prove to be just as insufficient in the future.

4 Feasibility

The feasibility of AGI is characterized by two central questions. The first, 'When will AGI be a reality?', addresses possible timelines for AGI development. The second question, 'How will AGI be implemented?' is about the technical details and ideas on approaches for AGI implementation. In the following, we present the most common answers to both questions. We also highlight how the ambiguous definition of AGI, which we discussed in Section 2, influences both of them.

4.1 Timelines

The most straightforward method to establish a timeline that describes the development of AGI is to survey experts and consolidate their individual estimates into an overall prognosis. This approach has been applied in several studies in recent years (Grace et al., 2025; Grace, Stein-Perlman, Weinstein-Raun, & Salvatier, 2022; Roser, 2023; Zhang et al., 2022). The surveyed experts typically include computer scientists as well as analysts, scientists of other AGI-related fields, and influential AI industry workers and leaders.

The main problem with using surveys to establish timeline estimates is that the expert's predictions heavily depend on their individual and subjective interpretation of the ambiguous AGI definition. While surveys try to combat this effect by phrasing the questions as clearly as possible, there is currently no universally accepted and unambiguous AGI definition to refer to. As a result, the experts' answers in most surveys are broadly distributed between the extremes of 'AGI has already been achieved' (Agüera y Arcas & Norvig, 2023; Al-Sibai, 2024) and 'AGI is conceptually impossible to implement' (Dreyfus, 1992; Fjelland, 2020). An example for a timeline predicted by an expert survey is shown in Figure 1.

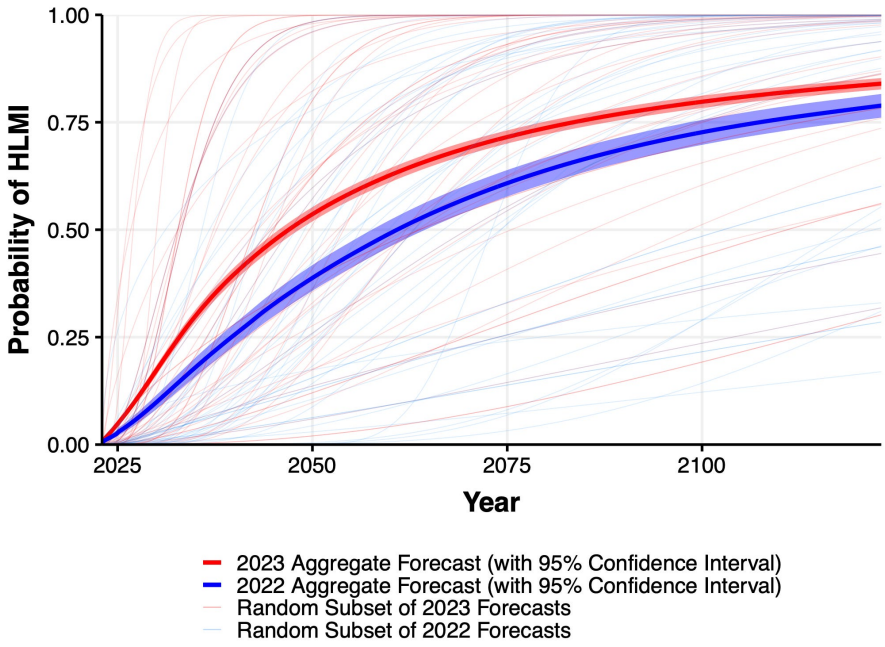


Figure 1: Estimated timeline of AGI development (here called HLMI for high-level machine intelligence) compiled from 2,778 experts surveyed in 2023. The experts selected for this survey are AI researchers with publications at highly regarded conferences (Grace et al., 2025).

Besides the high variance in expert predictions, such timelines also have the problem that expert surveys are an imprecise form of development prognosis in general. As we exemplified with some examples of incorrect predictions of renowned AI experts in Section 1, individual expert predictions are unreliable. And there is no definitive evidence that combining many expert estimations in large-scale surveys improves the prediction reliability. Besides their individual and subjective understanding of AGI, experts' survey answers are likely also influenced by the current AI development pace, future funding prospects, and public expectations. The history of the field of AI, with multiple AI winters and phases of AI booms, has shown that these conditions tend to change regularly, often quite suddenly and drastically.

4.2 Technical Approaches

Because there are diverse understandings of AGI, there are also many ideas about how AGI could be implemented on a technical level. In the following, we present the currently most commonly pursued approach of scaling up neural-

network-based AI models, as well as the most prominent alternative approach: adhering closer to biological analogs.

4.2.1 Scaling Neural Network Architectures

Most state-of-the-art AI systems are based on neural network architectures. In the context of AGI, transformer-based LLMs are currently most debated to potentially represent a predecessor of future AGI. The predominant way of advancing these systems is to scale up the underlying networks. In this case, scaling means using larger networks with more parameters, which in turn constitutes a need for more training data and increasingly large computational resources. The increase in size of LLMs from 2018 to 2022 is displayed in Figure 2. Besides increasing network scale, advancements in network architectures and training methods are also an important part of further developing AI systems. Recent examples of these include the aforementioned transformer architecture (Vaswani et al., 2017), as well as the use of reinforcement learning from human feedback (Ouyang et al., 2022) and forward-only gradient calculation (Baydin, Pearlmutter, Syme, Wood, & Torr, 2022).

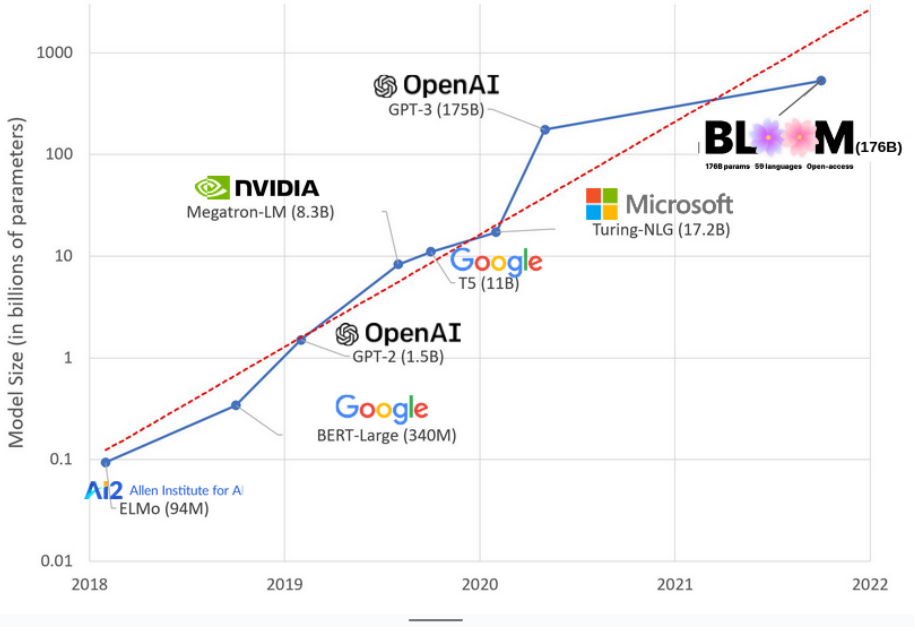


Figure 2: Parameter numbers of state-of-the-art LLMs between 2018 and 2022. Since the parameter axis is log-scaled, the plot represents a roughly exponential increase over time (Luccioni, 2023).

The predominant problem with scaling up neural networks is the limitation

in the availability of training data and computing resources (Hidary, 2025). As network parameter numbers scale roughly exponentially (cf. Figure 2), high-quality training data starts to run out, and the building up of compute power lags behind. The root cause of this high training data requirement is the relatively low data efficiency of neural networks (Huber, Mast, & Wichmann, 2024). Compared to humans, AI systems usually require more training data to generalize well for a given task.

From the perspective of AGI development, the problems associated with scaling neural networks are not the main concern with this approach. It is, instead, the lack of concrete evidence that scaling neural-network-based AI systems can or will eventually lead to AGI. While large neural networks are the approach that currently yields the most advanced AI systems, there is no guarantee that this development will continue, and it may not lead to AGI becoming a reality.

4.2.2 Bioinspired AI

One of the most investigated alternatives to artificial neural networks as the basis of modern AI systems is architectures that more closely resemble the biological neural foundations of human intelligence. While the components of artificial neural networks are referred to as neurons, they are more of a mathematical framework for function approximation than a detailed replication of neuron cells (Sarfraz, Arani, & Zonooz, 2023). A more realistic emulation of biological neuron structures on both the hardware and the architectural level has the potential to provide human-level learning efficiency and might also be able to replicate human-like intelligence in the future.

The most common architecture in bioinspired, also referred to as neuromorphic, computing is spiking neural networks (Maass, 1997). Like pulses in biological neuron systems, these networks do not encode signals in the amplitude of neuron activations but in the timing and or frequency of signal spikes. To emulate the behavior of biological neurons efficiently, specialized neuromorphic computing hardware is being developed. An example of this is the spiking neural network architecture (SpiNNaker), which was used to construct a supercomputer with over one million ARM cores specialized for neuron emulation (Furber, Galluppi, Temple, & Plana, 2014). Its predecessor, SpiNNcloud, deployed at the TU Dresden, uses more than five million cores to emulate over 5.2 billion neurons, which makes it capable of emulating a single digit percentage of the neurons in a human brain (Gonzalez et al., 2023).

Advancements in the computational power of neuromorphic hardware make it seem plausible that the emulation of an entire brain on the neuron level might become possible in the future. But the main problem with bioinspired AI lies in the network architecture. Efforts like the Human Brain Project work on capturing the brain's structure down to individual connections between neurons (European Commission and Directorate-General for Communications Net-

works, Content and Technology, 2024). But scanning techniques and resolutions are currently insufficient to yield a complete brain-sized network to emulate (Zanichelli, Schons, Freeman, Shiu, & Arkhipov, 2025). Additionally, it is currently undetermined whether a detailed replication of a human brain is sufficient for replicating human intelligence and therefore achieving AGI (Russell & Norvig, 2021, pp.1798-1801). Even though the neuromorphic approach is more direct than scaling artificial neural networks, the monistic position of physicalism is not undisputed in philosophy. Positions like mind–body dualism dispute that physical replication of the human brain is sufficient to replicate human intelligence.

5 Implications

Talking about the implications of AGI necessarily involves making predictions of the future. This is, as in all other scientific areas, a complex and difficult task with mostly unreliable or unspecific outcomes. To make a prediction, assumptions for several currently unanswered questions have to be made. We have covered some of the most important questions in the previous chapters. They include 'What is intelligence?', and 'What properties does AGI have?' (cf. Sections 2 and 3), as well as 'How does AGI work (on a technical level)?' and 'When will AGI become a reality?' (cf. Section 4). In the respective chapters, we also outlined how the ambiguous definition of AGI impacts these questions, leading to vague and diverse possible answers.

Since making predictions about the implications of AGI requires assuming answers to these questions, the ambiguity rooted in the vague definition of AGI propagates through increasingly concrete sub-fields of AGI and accumulates in the resulting predictions. The impact of the ambiguous AGI definition is therefore stronger the more 'layers' are added on top of it, and has the highest impact in speculations on future AGI impacts. Minimizing the number of assumptions made to come to a prediction can limit the resulting uncertainty and increase the probability of the resulting prediction being correct. But at the same time, this approach produces very general and unspecific predictions that often appear obvious and offer little novel insight. The alternative approach is to make more assumptions, which leads to more specific and interesting predictions. The downside of this approach is that the chance of some assumptions and, therefore, the resulting prediction being incorrect is high.

Since the first option is of little value to the reader, and the other is approaching the domain of science fiction, we do not discuss any of the numerous existing predictions about the impacts of AGI in detail. Instead, we want to point out how the ambiguous definition of AGI makes it significantly harder to make predictions about AGI compared to other technological topics. Therefore, the first step in analyzing or discussing any statement on the implications of AGI should be to identify the underlying assumptions encoded in that statement.

6 Conclusion and Outlook

We have presented an overview of the current state and challenges of the development of AGI, as well as the discussion about it. We highlighted that intelligence is not well understood and explained that this makes the definition of AGI, which is based on the concept of intelligence, ambiguous. Based on this insight, we discussed how the ambiguity of the AGI definition allows for individual and subjective interpretations and understandings of AGI. These diverse ideas of what AGI is, in turn, lead to a broad range of AGI benchmarking approaches, timeline estimations, technical methods to implement AGI, and predictions on AGI's future impacts.

The past few years have brought several impressive and impactful developments in the field of AI. However, the history of AI as a discipline has so far always been characterized by alternating phases of AI booms and AI winters. It is therefore important to acknowledge that there is no guarantee for a continuous development towards AGI from this point. A fact that is underscored by the inaccurate predictions we presented in Section 1. Additionally, the ambiguous foundation on which the concept of AGI is built makes targeted development towards AGI all but impossible. Without an unambiguous understanding of what intelligence, and therefore AGI, is, the development of true AGI systems would be rather incidental. And the chances of achieving and then recognizing an AGI as such are low. This means that while most of the focus in the discussion about AGI is on technical aspects in the field of computer science at the moment, new insights about the concept of intelligence from fields like neuroscience or psychology might prove to be more impactful on AGI development. To maximize the chances of success in the purposeful creation of AGI, the main effort needs to be directed at the currently insufficiently clear conceptual foundations before turning to details of technical implementation.

Considering that the realization of AGI might not be imminent, any discussion about it, either on a technical level or about its implications, could be seen as premature. It is, however, important to recognize that the discussion about any technological and societal advancement is best begun before its realization. The more impactful a development could be, the more important it becomes to start a discussion about it sooner rather than later. At the same time, an early consideration of AGI should not distract from the discussion on and efforts to solve problems caused by current AI developments.

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